

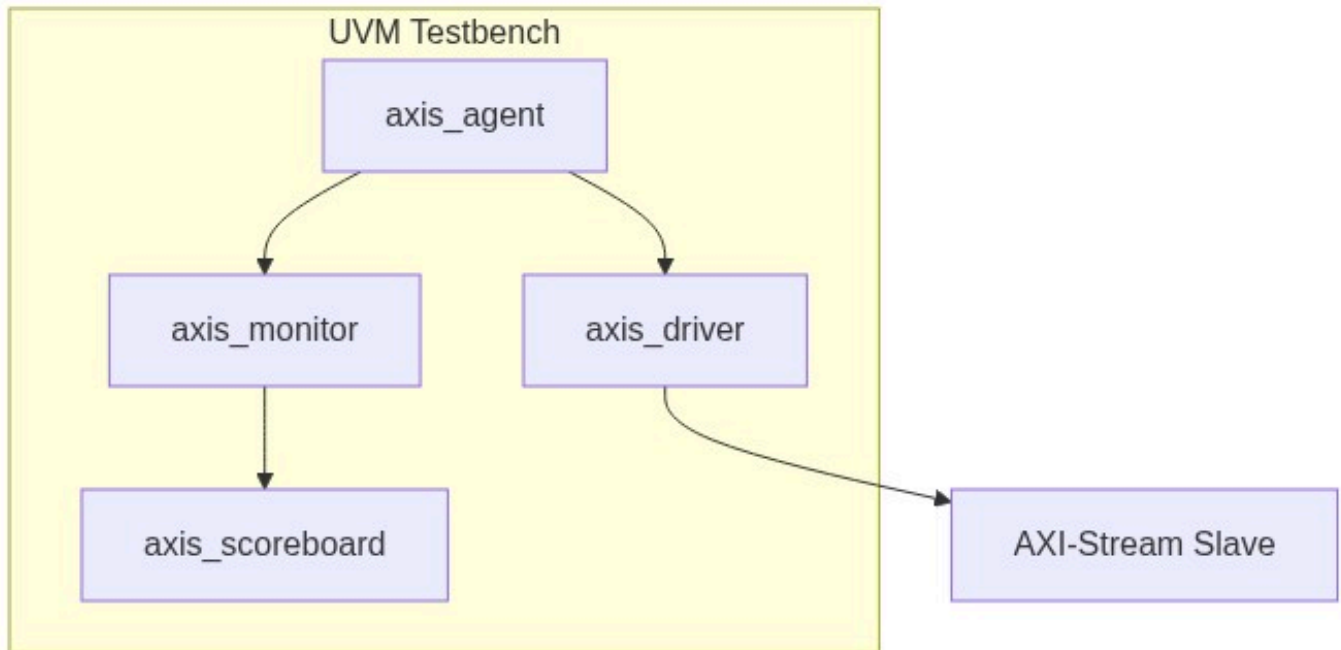
Verification Plan Report

Project: AXI-Stream Enterprise Mastery

Version: 1.9

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_HS_01	Handshake (Stability)	Master Data Stability	Master shall not change TDATA/TSTRB/TKEEP/TLAST/TID/TDEST/TUSER while TVALID is HIGH and TREADY is LOW.
REQ_HS_02	Handshake (Timing)	Valid Independence	Master must not wait for TREADY to be asserted before asserting TVALID.
REQ_RST_01	Reset (Suppression)	Reset TVALID LOW	TVALID must be driven LOW at least one cycle before reset deassertion and remain LOW during reset.
REQ_RST_02	Reset (Recovery)	Post-Reset Latency	Master must wait at least 1 cycle after ARESETn goes HIGH before driving TVALID.
REQ_PKT_01	Packet (ID)	TID Consistency	TID must remain constant within a packet (between TLAST pulses).
REQ_PKT_02	Packet (DEST)	TDEST Consistency	TDEST must remain constant within a packet (between TLAST pulses).

Verification Plan Report

REQ_QUAL_01	Qualifiers (TKEEP)	Null Identification	TKEEP bits identify which data bytes are processed.
REQ_QUAL_02	Qualifiers (TSTRB)	Strobe Logic	TSTRB identifies data vs position bytes.
REQ_PWR_01	Power (AXI5)	Wakeup Activity	TWAKEUP must indicate activity and be synchronous to ACLK.
REQ_PWR_02	Power (AXI5)	Wakeup Protocol	If TVALID and TWAKEUP are concurrent, TWAKEUP must hold until handshake.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_HS_01	Valid-Before-Ready: Master drives TVALID=1, Slave stays 0 for 5 cycles. Verify stability.	REQ_HS_01
TC_HS_02	Ready-Before-Valid: Slave drives TREADY=1. Master asserts valid. verify 1 cycle handshake.	REQ_HS_02
TC_HS_03	Back-to-Back: Mastery throughput test with infinite 1-cycle beats.	REQ_HS_01
TC_HS_04	Master Stall: Master deasserts TVALID mid-packet. Verify Slave waits.	REQ_HS_02
TC_RST_01	Reset Suppression: Verify TVALID goes LOW 0 cycles after ARESETn assertion.	REQ_RST_01
TC_RST_02	Post-Reset Timing: Attempt TVALID=1 on same cycle as ARESETn=1. Fail check.	REQ_RST_02
TC_PKT_01	Packet Length 1: Single beat with TLAST=1.	REQ_PKT_01
TC_PKT_02	Packet Length 1024: Long packet with TID/TDEST stable throughout.	REQ_PKT_01, REQ_PKT_02
TC_PKT_03	Null Packet: TLAST=1 but all TKEEP bits are 0.	REQ_QUAL_01
TC_PKT_04	ID Interleaving: Packet 1 (TID=0), Packet 2 (TID=1) switching.	REQ_PKT_01
TC_PKT_05	Illegal ID Change: Change TID while TLAST=0. Verify assertion.	REQ_PKT_01
TC_PKT_06	Illegal DEST Change: Change TDEST while TLAST=0. Verify assertion.	REQ_PKT_02
TC_QUAL_01	Sparse TSTRB: Transfer where only odd bytes are marked as data.	REQ_QUAL_02
TC_QUAL_02	Sparse TKEEP: Transfer where middle bytes are nullified.	REQ_QUAL_01
TC_QUAL_03	TSTRB/TKEEP Logic: Verify TSTRB=TKEEP default when TSTRB is absent.	REQ_QUAL_02
TC_PWR_01	Wakeup Pre-assertion: TWAKEUP asserted 1 cycle before TVALID.	REQ_PWR_01
TC_PWR_02	Wakeup Concurrency: TWAKEUP and TVALID asserted same cycle.	REQ_PWR_02
TC_PWR_03	Early Wakeup Drop: Drop TWAKEUP before TREADY is HIGH. Verify failure.	REQ_PWR_02
TC_SIDE_01	TUSER Stability: Toggle TUSER bits during TREADY stall. Verify violation.	REQ_HS_01

Verification Plan Report

TC_SIDE_02	TUSER Large Width: Verify TUSER_WIDTH > 32 performance and data integrity.	REQ_HS_01
TC_STRESS_01	Random Pressure: Random TREADY and random TVALID gaps for 1000 transfers.	REQ_HS_02
TC_STRESS_02	Heavy Interleaving: Randomly switch TID every packet with zero-cycle gaps.	REQ_PKT_01
TC_STRESS_03	Configuration Stress: Change TDATA_WIDTH and re-verify handshake logic.	REQ_HS_01
TC_VAL_01	Valid Toggle: Toggle TVALID every cycle to force handshake delays.	REQ_HS_01
TC_VAL_02	Ready Hold: Hold TREADY LOW for 100 cycles. Verify zero data movement.	REQ_HS_01
TC_MISC_01	Arbitration Change: Verify TLAST assists in arbitration weighting.	REQ_PKT_02
TC_MISC_02	Buffer Flush: Assert TLAST with zero data to push through buffers.	REQ_PKT_02
TC_CORNER_01	Max Width Data: Verification of 1024-bit data path throughput.	REQ_HS_01
TC_CORNER_02	Minimal Config: TDATA omitted. Verification of pure control flow.	REQ_HS_02
TC_CORNER_03	Position-only Beats: Beats containing only position bytes (TSTRB=1, TKEEP=0).	REQ_QUAL_02
TC_FINAL_01	End-to-End Regression: Run all 31 previous cases in sequence.	REQ_HS_01
TC_FINAL_02	Protocol Wrap: Comprehensive status check post-simulation.	REQ_HS_02

4. Functional Coverage Groups

Group Name	Description
cg_handshake	Cross coverage of TVALID/TREADY.
cg_packet	TLAST transitions and ID switches.

5. Interface Signals

Signal Name	Direction	Width	Description
TDATA	Master	TD_W	Data
TVALID	Master	1	Valid
TREADY	Slave	1	Ready
TLAST	Master	1	Packet End
TID	Master	TID_W	Stream ID
TDEST	Master	TDEST_W	Dest ID
TKEEP	Master	TD_W/8	Null
TSTRB	Master	TD_W/8	Strobe

Verification Plan Report

6. Verification Environment Structure

```
verif/axis/  
|-- sv/  
|   |-- axis_driver.sv  
|   |-- axis_monitor.sv  
|   |-- axis_agent.sv  
|   |-- axis_if.sv  
|-- tests/  
    |-- base_test.sv  
    |-- stress_test.sv
```

7. SystemVerilog Implementation

Monitor Protocol Check

```
if (vif.TVALID && vif.TREADY) begin  
    // Perform sample  
    tr.data = vif.TDATA;  
    tr.last = vif.TLAST;  
end
```